

Generative Models in Finance

Week 3: LLMs in Finance and Investment Management

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[Reference:](#) Y. Kong, Y. Nie, X. Dong, J. M. Mulvey, H. V. Poor, Q. Wen & S. Zohren, *Large Language Models for Financial and Investment Management: Applications and Benchmarks*, Journal of Portfolio Management, 2024.

The Unstructured Data Frontier

- Quantitative finance has traditionally relied on **structured data**: prices, volumes, balance sheet items — numbers in rows and columns
- Over 80% of financial information is **unstructured text**: regulatory filings, earnings transcripts, analyst reports, central bank statements, news, social media
- Until recently, this text was largely inaccessible to systematic strategies
- **LLMs change this**. The rest of this lecture examines five ways they do it — and where they fall short

What Is Sentiment Analysis?

- **Sentiment analysis** assigns a score (positive, negative, or neutral) to a piece of text, quantifying its opinion or tone
- **Financial application:** given a news headline, an earnings transcript, or a central bank statement, estimate how *positive* or *negative* the market-relevant content is
- **Why it matters:** if sentiment can be measured *before* the market fully reacts, it becomes a trading signal
- **Core sub-tasks** in the literature:
 - ▶ Scoring news headlines and social media posts
 - ▶ Analysing tone in corporate disclosures and earnings calls
 - ▶ Interpreting central bank communications (FOMC minutes, ECB statements)

Before LLMs: Dictionary and Embedding Methods

- **Lexicon-based methods:** count occurrences of words from predefined positive/negative dictionaries
 - ▶ The **Loughran–McDonald (2011)** word list is finance-specific: “liability” is negative in everyday English but neutral in a financial filing
 - ▶ Simple and interpretable, but cannot handle negation (“not profitable”), sarcasm, or context-dependent meaning
- **Machine learning methods:** train classifiers (SVM, Naïve Bayes, random forests) on labelled sentiment datasets
 - ▶ More flexible, but require large labelled corpora for each new domain
- **Embedding methods** (Word2Vec, GloVe, Doc2Vec): represent words as vectors in $\mathbb{R}^d$ where semantic similarity $\approx$ cosine similarity
 - ▶ Capture context better than word counts, but produce *static* embeddings — the same vector regardless of surrounding words

LLMs for Sentiment: What Changed

- Pre-trained language models produce **contextualised embeddings**: every token's representation depends on its full surrounding context
- This means the model can distinguish “bank” (financial institution) from “bank” (river bank), and recognise that “not profitable” is negative even though “profitable” alone is positive
- **Four advantages** over prior methods:
 - ① Handle informal language, emojis, sarcasm, and sector-specific jargon
 - ② Process multimodal inputs (text + audio from earnings calls)
 - ③ Analyse long documents end-to-end, not just isolated sentences
 - ④ Show greater resilience to adversarial text manipulation

Finance-Specific vs. General-Purpose LLMs

- **FinBERT** (Araci 2019): takes the BERT architecture (110M parameters) and fine-tunes it on a large corpus of financial news. It learns that “liability,” “short,” and “hedge” have domain-specific meanings. Widely used as a baseline for financial sentiment classification
- **BloombergGPT** (Wu et al. 2023): a 50-billion-parameter model *pre-trained from scratch* on a mix of general text and Bloomberg's proprietary financial data (363B tokens of financial text). Outperforms general-purpose models of comparable size on financial NLP tasks, while remaining competitive on general benchmarks
- **FLANG** (Shah et al. 2022): BERT and ELECTRA variants pre-trained with a financial-domain masking strategy that targets financial keywords and phrases
- **General-purpose models** (GPT-4, Claude, LLaMA): not trained specifically on financial data, but can be used zero-shot or with prompting. Often competitive with or superior to fine-tuned models, especially on tasks requiring broad world knowledge
- **Open question**: when does domain-specific pre-training justify its cost? The answer depends on the task, the data, and how quickly frontier general models improve

Wu, S. et al. (2023). BloombergGPT: A Large Language Model for Finance. arXiv:2303.17564.
Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-Trained Language Models. arXiv:1908.10063.
Shah, R.S. et al. (2022). When FLUE Meets FLANG: Benchmarks and Large Pre-Trained Language Models for Financial Domain. arXiv:2211.00083.

LLM Sentiment Predicts Returns (I)

- **Lopez-Lira & Tang (2023):** give GPT-4 a single news headline, ask it to score sentiment on a numerical scale, then test whether this score predicts the *next day's* cross-sectional stock return
- **Result:** statistically significant predictive power, surviving standard controls
- The effect is strongest for **small-cap stocks** (less liquid, less analyst coverage) and after **negative news** — consistent with theories of slow information diffusion and limits to arbitrage
- GPT-4 outperforms all baselines on the same data: GPT-2, BERT, and the Loughran–McDonald dictionary
- **Interpretation:** the LLM captures nuances in language (irony, hedging, context) that simpler methods miss, and this extra information has not yet been priced in

LLM Sentiment Predicts Returns (II)

- [Chen, Kelly & Xiu \(2022\)](#): use BERT, RoBERTa, and OPT to score the sentiment of news articles, then sort stocks into portfolios by sentiment
- Portfolios based on LLM sentiment achieve [higher Sharpe ratios](#) (excess return divided by volatility) than those based on Word2Vec embeddings
- Crucially, news information is incorporated into stock prices with a [delay](#) — creating a short-term window where the LLM's reading of the news has not yet been reflected in prices
- [Steinert & Altmann \(2023\)](#): apply GPT-4 sentiment analysis to StockTwits microblog posts. A sentiment-based strategy for Apple and Tesla significantly outperforms buy-and-hold
- [Su, Mulvey & Poor \(2022\)](#): go beyond return prediction — they extract *covariance* estimates from Twitter using BERT, and feed these into mean-variance portfolio optimisation. The text-based portfolios outperform during the COVID-19 crash

Chen, Y., Kelly, B.T. & Xiu, D. (2022). Expected Returns and Large Language Models. SSRN 4416687.

Steinert, R. & Altmann, S. (2023). Linking Microblogging Sentiments to Stock Price Movement. arXiv:2308.16771.

Su, D.-J., Mulvey, J.M. & Poor, H.V. (2022). Improving Portfolio Performance Via NLP. *J. Financial Data Science* 4(2): 37–49.

The Weakest Claim: Robustness (I)

- **Adversarial fragility:** Leippold (2023) tests a simple attack on GPT-3 financial sentiment: take a sentence, replace a few words with synonyms (e.g. “revenue grew” → “revenue expanded”), and re-query the model
- **Result:** these minimal edits — which do not change the meaning — can **flip the sentiment score** from positive to negative or vice versa
- By contrast, the Loughran–McDonald dictionary is *immune* to this attack: synonyms map to the same polarity. The sophistication of LLMs introduces a new attack surface that simpler methods do not have
- **Implication for trading:** if a competitor or bad actor can manipulate the text inputs to your LLM-based strategy (e.g. by crafting press releases or social media posts), the resulting trading signals can be systematically corrupted
- This is not hypothetical: Cao et al. (2023) document that corporations are already adjusting the language of their disclosures in response to the knowledge that AI systems are reading them

Leippold, M. (2023). Sentiment Spin: Attacking Financial Sentiment with GPT-3. *Finance Research Letters* 55: 103957.
Cao, S. et al. (2023). How to Talk When a Machine Is Listening: Corporate Disclosure in the Age of AI. *Rev. Financial Studies* 36(9): 3603–3642.

The Weakest Claim: Robustness (II)

- **Multimodal gaps:** Curti & Kazinnik (2023) study FOMC press conferences and find that *nonverbal* cues (facial expressions, tone of voice) carry significant information about monetary policy. Text-only LLMs miss this entirely
- **Social media noise:** platforms like Twitter/X and StockTwits contain misinformation, sarcasm, and coordinated manipulation campaigns. LLMs trained on general text may not distinguish genuine sentiment from noise or deliberate deception
- **Out-of-sample fragility:** most sentiment-return studies test on 1–3 years of US equity data. Whether the signal persists across market regimes (bull vs. bear), asset classes (bonds, FX, commodities), and geographies remains largely untested
- **Signal decay:** if many participants deploy similar LLM sentiment strategies, the arbitrage opportunity should shrink over time as the market incorporates the information faster. There is no long-run evidence on whether these signals are durable
- **Verdict:** LLM sentiment is a promising *input* to a broader quantitative model, not a standalone trading strategy. Robustness testing — across time, assets, and adversarial conditions — is essential before deployment

Can Language Models Understand Time Series?

- LLMs are trained on *text*. A financial time series is a sequence of *numbers*. At first glance, these have nothing in common
- But both are **sequential data**: a sentence is a sequence of tokens, a price series is a sequence of values. The transformer architecture processes sequences — it does not inherently care whether the tokens represent words or digits
- **Key insight**: the web text that LLMs are pre-trained on contains vast amounts of numerical sequences — tables of stock prices, macroeconomic data in Wikipedia, financial statements in HTML. The model has already seen numbers in temporal context

Zero-Shot Time Series Forecasting

- [Gruver et al. \(2024\)](#) test this directly: take a pre-trained LLM (GPT-2, LLaMA), give it *no* additional training on time series, and ask it to forecast
- **Method:** convert a numerical series into a string of digit tokens (e.g. the sequence [1423.5, 1418.2, 1425.7] becomes the text “1423.5, 1418.2, 1425.7”) and prompt the model to continue
- **Result:** the model generates plausible continuations that capture trends, seasonality, and mean-reversion — *zero-shot*, with no fine-tuning
- **An alternative approach — Time-LLM** (Jin, Wang et al. 2023): instead of converting numbers to text, *reprogram* time series data into the LLM's embedding space using a learned projection. This preserves numerical precision and achieves state-of-the-art forecasting on standard benchmarks
- **Broader direction:** these results have sparked work on [foundation models for time series](#) — large pre-trained models specifically designed for temporal data, borrowing the transformer architecture from the LLM paradigm

Foundation Models for Time Series

- The success of zero-shot forecasting has motivated building **purpose-built foundation models** for time series — trained from scratch on massive temporal datasets, not repurposed from language
- **TimesFM** (Das et al. 2024, Google): a decoder-only transformer pre-trained on 100 billion real-world time points. Uses patching (grouping consecutive observations into tokens) to capture local structure. Performs zero-shot forecasting competitive with supervised models trained per-dataset
- **Chronos** (Ansari et al. 2024, Amazon): repurposes the T5 language model architecture for time series by tokenising numerical values through scaling and quantisation. Trained on both real and synthetic time series data
- **Moirai** (Woo et al. 2024, Salesforce): an encoder-only transformer that outputs *probabilistic* forecasts using a mixture of distributions, providing flexible prediction intervals rather than just point estimates
- **The analogy to NLP is direct**: just as GPT was pre-trained once and applied to many text tasks, these models are pre-trained once on diverse time series and applied zero-shot to new forecasting problems

Das, A. et al. (2024). A Decoder-Only Foundation Model for Time-Series Forecasting. *ICML* 2024.
Ansari, A.F. et al. (2024). Chronos: Learning the Language of Time Series. arXiv:2403.07815.
Woo, G. et al. (2024). Unified Training of Universal Time Series Forecasting Transformers. *ICML* 2024.

The Weakest Claim: LLMs as Standalone Forecasters (I)

- Xie, Han, Lai et al. (2023) conduct a systematic zero-shot evaluation of ChatGPT on multimodal stock movement prediction (combining text, price data, and images)
- Result: ChatGPT underperforms both traditional ML models (gradient-boosted trees, SVMs) and state-of-the-art deep learning models that were trained on the same data
- Why? Two fundamental problems:
 - ▶ Numerical precision: tokenising “1423.50” as a string of characters loses the notion that this is a number close to 1423 and far from 14. The model has no native understanding of numerical magnitude
 - ▶ Distributional shift: financial time series are non-stationary — their statistical properties change over time. An LLM pre-trained on historical web text has no mechanism to adapt to a regime it has never seen
- Even the dedicated foundation models (TimesFM, Chronos, Moirai) are evaluated primarily on *general* time series benchmarks (weather, electricity, traffic). Their performance on *financial* series — which are noisier, more adversarial, and lower signal-to-noise — is less established

The Weakest Claim: LLMs as Standalone Forecasters (II)

- **The efficient market objection:** if a publicly available LLM can predict returns from publicly available news, the signal should be arbitrated away rapidly. The persistence of such signals would imply a market inefficiency that is hard to reconcile with decades of asset pricing research
- **Confounds in backtests:** many studies (e.g. Yu et al. 2023, Chen et al. 2023) use LLMs trained *after* the test period — the model may have memorised events or patterns from its training data that overlap with the backtest window (look-ahead contamination)
- **Reproducibility:** commercial LLM APIs (GPT-3.5, GPT-4) change silently over time. A study run in January 2023 may not reproduce in June 2023 if the underlying model was updated. This makes rigorous out-of-sample testing difficult
- **Verdict:** LLMs are most convincing as **feature extractors** — generating text-derived signals (sentiment, entity relationships, summaries) that feed into classical quantitative models (regression, portfolio optimisation). Using an LLM as a direct price forecaster remains speculative and under-evidenced

What Is an LLM Agent?

- A standard LLM is **reactive**: you give it a prompt, it returns a response, the interaction ends
- An **LLM agent** adds three capabilities that turn it into an autonomous system:
 - 1 **Tool use**: the model can call external functions — fetch a stock price, query a database, execute a trade via an API, run Python code
 - 2 **Planning**: the model breaks a complex goal into sub-steps and executes them sequentially, deciding at each step what to do next based on prior results
 - 3 **Memory**: the model maintains persistent state across interactions — it remembers what it did yesterday and what the outcome was
- The combination means an agent can operate in a **loop**: perceive the environment → reason about what to do → act → observe the result → repeat
- **In finance**, this loop becomes: read today's market data and news → form a trading thesis → place an order → observe the P&L → update the strategy

LLM Trading Agents: Architecture and Memory

- The simplest trading agent: prompt an LLM with today's news and prices, ask “buy or sell?”. **Problem:** no memory — each query starts from scratch, no learning from past trades
- **FINMEM** (Yu et al. 2024): gives the agent working memory (current context) + layered long-term memory (past trades, outcomes, lessons) prioritised by relevance and recency. The agent evolves its strategy through continuous learning
- **FinAgent** (Zhang, Zhao et al. 2024): the first *multimodal* trading agent — integrates text, numerical data, and candlestick chart images via GPT-4V. Outperforms FINMEM by adding visual pattern recognition
- **QuantAgent** (Wang, Yuan et al. 2024): a self-improving agent with two loops — an inner loop refines strategies via a knowledge base, an outer loop validates on real market data. Discovers trading signals autonomously
- **Alpha-GPT 2.0** (Yuan et al. 2024): human-in-the-loop — a trader proposes alpha ideas, the agent formalises, backtests, and refines them iteratively

Yu, Y.H. et al. (2024). FinMem. *AAAI Symposium Series*, Vol. 3, pp. 595–597.

Zhang, W. et al. (2024). FinAgent: A Multimodal Foundation Agent for Financial Trading. arXiv:2402.18485.

Wang, S. et al. (2024). QuantAgent. arXiv:2402.03755. Yuan, H. et al. (2024). Alpha-GPT 2.0. arXiv:2402.09746.

Multi-Agent Systems and Market Simulation

- **TradingAgents** (Xiao et al. 2024): mimics the structure of a real trading firm — specialised LLM agents act as fundamental analysts, sentiment analysts, technical analysts, and traders with different risk profiles. Agents debate and synthesise reports before the trader decides. Outperforms single-agent baselines on US equities (Jun–Nov 2024)
- **StockAgent** (Zhang, Liu et al. 2024): populates a simulated market with LLM agents. GPT agents trade independently; Gemini agents follow trends. Removing information channels (bulletin boards) significantly alters dynamics — revealing how information flow drives price formation
- **“Homo Silicus”** (Horton 2023): LLM agents endowed with behavioural-economics preferences simulate human economic decisions. These “silicon subjects” can replace costly human experiments in social science
- **No strong evidence** that any LLM agent consistently generates alpha in *live* markets. Most results are simulated or backtested
- **Verdict**: multi-agent systems are powerful for *research* and *augmentation*. Fully autonomous LLM trading remains speculative

Challenge 1: Hallucination

- **What is hallucination?** An LLM generates text that is fluent, confident, and **factually wrong** — not because it is “lying,” but because it is a statistical model of language that optimises for plausibility, not truth
- **Why is this dangerous in finance?** A hallucinated revenue figure looks identical to a real one. A fabricated regulatory citation could pass a compliance check. Unlike a search engine that returns “no results,” an LLM *always* produces an answer
- **Concrete example:** ask an LLM “What was Apple’s Q3 2023 revenue?” — it may produce a precise number (e.g. \$81.8B) that is close to the real figure (\$81.8B) or slightly off (\$82.1B), with no way to tell which from the output alone
- **Mitigation strategies:**
 - ▶ *Retrieval-Augmented Generation (RAG)*: force the model to retrieve and cite source documents before answering — the answer is grounded in real data
 - ▶ *Code-based reasoning*: for numerical questions, have the LLM write Python code that computes the answer rather than generating the number directly
 - ▶ *Multi-agent verification*: a second LLM checks the first’s output against retrieved sources

Challenge 2: Bias and Fairness

- **What is model bias?** LLMs learn statistical patterns from their training data. If certain stocks, sectors, or viewpoints are over-represented in the training corpus, the model's outputs will reflect those biases
- **Evidence in finance:** Chuang & Yang (2022) probe BERT and FinBERT with prompts like “[MASK] is a good investment” and find the models systematically prefer certain stocks (e.g. Tesla) over others (e.g. Ford). An investment system built on these embeddings would inherit these biases as systematic tilts
- **Fairness in advice:** Lakkaraju et al. (2023) test ChatGPT and Bard on personal financial questions across English, African American Vernacular English, and Telugu. The models give **inconsistent recommendations across demographics** — sometimes differing on whether to recommend a savings account vs. a CD based solely on the dialect of the query
- **Over-reliance:** Niszczoła & Abbas (2023) find GPT-4 scores near-perfect on financial literacy tests. But users with *lower* financial knowledge rely on it *more heavily* — precisely the population most vulnerable to errors

Chuang, C. & Yang, Y. (2022). Buy Tesla, Sell Ford. *ACL*, pp. 100–105.

Lakkaraju, K. et al. (2023). LLMs for Financial Advisement. *ACM Int. Conf. on AI in Finance*, pp. 100–107.

Niszczoła, P. & Abbas, S. (2023). GPT Has Become Financially Literate. *Finance Research Letters* 58: 104333.

Challenge 3: Regulation and Liability

- **The core question:** if an LLM gives financial advice and the client loses money, who is liable? The LLM provider? The firm that deployed it? The individual who acted on it?
- **Existing frameworks were not designed for AI:**
 - ▶ *Fiduciary duty* requires advisors to act in the client's best interest — but an LLM has no concept of “interest” and cannot be held to a legal standard
 - ▶ *Suitability requirements* mandate that advice be appropriate for the client's risk profile — but an LLM may not reliably distinguish a retiree from a day trader
 - ▶ *Explainability requirements* (e.g. MiFID II in the EU) demand that firms explain *why* a recommendation was made — chain-of-thought reasoning may or may not satisfy this
- **Cao et al. (2023)** document that corporations are already adapting their disclosures to be more “machine-readable” — a new form of strategic communication aimed at AI readers rather than human analysts
- **No jurisdiction has yet established clear rules** for AI-generated financial advice. This regulatory vacuum is a risk for both firms and consumers

Open Questions

- Can LLMs genuinely predict financial markets, or are they extracting sentiment signals that could be obtained more simply?
- When does domain-specific fine-tuning (FinBERT) outperform zero-shot prompting of a frontier model (GPT-4)?
- How do we defend LLM-based trading systems against adversarial manipulation of their text inputs?
- Can chain-of-thought reasoning satisfy regulators' explainability requirements?
- If many market participants deploy similar LLM agents, what happens to market stability?
- The most promising direction may be **hybrid systems**: LLM reasoning combined with classical portfolio optimisation, risk models, and rigorous statistical testing